Resampling

Same questions as the classical tests — but the computer builds the answer out of your own data instead of reading it off a theoretical curve.

WHAT IT IS

The classical tests were designed when arithmetic was done by hand. To avoid impossible calculations they assume a shape for the data — usually the normal curve — derive a formula from it, and read the answer out of a printed table. Resampling drops the assumption and uses brute force instead: take your own measurements and make thousands of copies of the study, each one reshuffled or redrawn at random. What varies across those copies is the uncertainty you were trying to quantify. The computer does in a second what the formulas were invented to avoid.

- **Nothing is assumed about the shape of the data.** Skewed, bounded, lumpy, ordinal — the method does not care, because it uses your distribution rather than an idealised one.
- **It works for statistics that have no formula.** Medians, ratios, differences between medians, the number needed to treat: if you can compute it once, you can resample it.
- **It suits small samples,** where the large-sample approximations behind the classical tests are least trustworthy — and small samples are what most clinical pilot work produces.
- **You can see it happen.** The null distribution appears as a histogram on the screen instead of hiding inside a table, which makes it far easier to explain to a co-author, a reviewer, or a student.
- **It costs no extra patients.** Resampling extracts more from the data you already have; it does not invent information, and it cannot rescue a study that was badly designed.

WHICH ONE ANSWERS WHICH QUESTION

YOUR QUESTION	METHOD	WHAT YOU REPORT
Is the difference more than chance?	`Permutation test`	a p-value
How precise is my estimate?	`Bootstrap`	a 95% confidence interval
How big must the real study be?	`Bootstrap a pilot`	a sample size, and its uncertainty

01 Permutation test

If the treatment did nothing, the labels are interchangeable. So write all twelve measurements on cards, forget which treatment each came from, deal six to each group at random, and note the difference in means. A few thousand deals give you the differences that chance alone produces — and your real difference is just one point on that picture.

```
# probing depth (mm) after two treatments, 6 patients each
a <- c(3.1, 2.8, 4.0, 3.6, 2.9, 3.4)
b <- c(2.4, 2.1, 3.0, 2.6, 2.2, 2.8)

obs  <- mean(a) - mean(b)
pool <- c(a, b)

diffs <- replicate(10000, {
  i <- sample(12, 6)                # deal the labels at random
  mean(pool[i]) - mean(pool[-i])
})

mean(abs(diffs) >= abs(obs))        # the p-value
hist(diffs); abline(v = obs, col = "red")
```

THE SHUFFLE LINE, UNPACKED

```
pool          # the 12 measurements, origin forgotten
i <- sample(12, 6)   # six *positions* out of twelve, no repeats
pool[i]              # the values sitting there  -> the new group A
pool[-i]             # minus = "all the others"  -> the new group B
```

The minus sign is R for *leave these out*, not for a negative element. Every deal uses the same twelve numbers — only the labels move, which is exactly what the null hypothesis claims.

Reads as: out of 10 000 random relabellings, this share was as extreme as what we saw. The histogram is the null hypothesis, drawn.

02 Bootstrap

You cannot repeat the study 10 000 times, so draw 10 000 new samples from your own data, with replacement. How much the answer moves is how uncertain it is.

```
# cost per treatment (CHF) — skewed, no formula for a median CI
x <- c(210, 340, 180, 1250, 260, 890, 300, 220, 640, 195)

meds <- replicate(10000, median(sample(x, 10, replace = TRUE)))

median(x)
quantile(meds, c(0.025, 0.975))     # 95% confidence interval
```

Reads as: the median cost with an interval around it — and nothing was assumed about the shape of the data.

03 Sample size from a pilot study

A pilot study is not there to prove anything. It is there to measure how much your outcome scatters, so you can work out how many patients the real study needs. And because the pilot is small, that scatter is itself uncertain — so bootstrap it.

```
# pilot: pocket depth reduction (mm) in 8 patients
pilot <- c(1.1, 0.4, 1.6, 0.9, 1.3, 0.2, 1.5, 0.8)

sd(pilot)

# patients per group needed to detect a 0.5 mm difference, 80% power
power.t.test(delta = 0.5, sd = sd(pilot), power = 0.8)$n

# how sure are we of that sd? resample the pilot
ns <- replicate(2000, {
  s <- sd(sample(pilot, 8, replace = TRUE))
  power.t.test(delta = 0.5, sd = s, power = 0.8)$n
})

quantile(ns, c(0.025, 0.975))
```

Reads as: the point estimate says about 17 patients per group — but eight pilot patients pin that down so loosely that the plausible range spans a factor of several. Plan for the upper end, not the middle. And take only the scatter from a pilot: its p-value and its effect size are noise.

RULES OF THUMB

1. **How many replicates?** At least 2000 for a p-value, 10 000 for a confidence interval. On any laptop this costs seconds.
2. **Resample the independent unit** — the patient, not the tooth, not the surface, not the visit. Take all of a patient's rows along together.
3. **Permutation and bootstrap answer different questions.** Permutation: is there a difference? Bootstrap: how big is it, and how sure are we? Most papers need both.
4. **Resampling relaxes distribution assumptions only.** It cannot repair selection bias, confounding, or a sample that is simply too small.